Substitution method

1

a $x=-8, y=8$

b $x=-5, y=3$

c $x=-8, y=-3$

d $x=-6, y=-3$

e $x=-3, y=5$

f $x=6, y=4$

g $x=9, y=5$

h $x=-2, y=9$

i $x=1, y=6$

j $x=8, y=4$

k $x=-5, y=-8$

l $x=8, y=1$

m $x=-\frac{4}{15}, y=\frac{13}{3}$

n $x=7, y=4$

2

a $x=\frac{4+7y}{3}$

b Infinitely many solutions.

c Dependent

3

a $y=-\frac{64}{3}-\frac{5x}{3}$

b $x=-8$

c $y=-8$

4

a $x=7, y=4$

b $x=-6, y=5$

c $x=6, y=-5$

d $x=2, y=3$

e $f=\frac{2}{5}, g=\frac{6}{5}$

f $q=\frac{4}{5}, p=\frac{1}{8}$

g $a=\frac{3}{2}, b=\frac{5}{2}$

h $v=\frac{2}{5}, t=\frac{5}{4}$

5 By substituting the ordered pair into each equation and seeing if it satisfies them.

6 Yes, it is a solution.

7 Yes, it is a solution.

8

a $x=4, y=2$

b Yes

c Yes

Applications

9

a $x=y+10$

b $x+y=28$

c $y=9$

d $x=19$

10

a $y=x+6$ **b** $x+y=22$ **c** $x=8$ **d** $y=14$

11

a $y=10x, y=7x+12$ **b** $x=4$

c $y=40$

12

a $y=5x, y=9x-32$ **b** $x=8$

c $y=40$

13

a $x+y=3000, 0.06x+0.08y=204$ **b** $x=\$1800$ and $y=\$1200$

14

a $2f+3g=10.08$ and $5f+7g=\$24.08$ **b** $f=\$1.68$ and $g=\$2.24$

15

a $y=4x-190\,000,\ y=x+560\,000$ **b** $x=250\,000$ and $y=810\,000$

16

a $x+10y=105$ and $x+15y=155$ **b** $x=5,\ y=10$

c $\$255$

17 Each racket is $\$500$ and each ball is $\$50$.

18 15 pieces of $10c$ coins and 45 pieces of $5c$ coins

19 35 hotdogs and 52 sodas.